

Scrolling terminal output in tmx

This build is configured so you can scroll back through terminal output **from any device** — a desktop mouse, a laptop trackpad, or a phone (Termux on Android) — and it works the same inside full-screen TUIs such as the Claude Code TUI.

All of this is set up automatically by `bash scripts/setup.sh`. The settings live in `scripts/tmux.conf.template`, which the `tmx` wrapper loads on every `tmx new`.

What you get

Setting	Value	Why
<code>history-limit</code>	<code>50000</code> lines	Output history survives — you can scroll a long way back.
<code>mode-keys</code>	<code>vi</code>	vi-style navigation in copy-mode.
<code>mouse</code>	off (default)	The terminal owns the mouse, so native click-drag selection (including multi-line), right-click → Copy, and the terminal's own scrollbar / wheel scroll all work — identically on Linux and macOS, on every emulator (iTerm2, Terminal.app, GNOME Terminal, WezTerm, ...).
<code>prefix m</code>	<code>toggle mouse on</code>	On demand: enable tmux's own mouse so the wheel drives tmux scrollbar inside full-screen TUIs (Claude Code / HelixCode), plus tmux drag-select that copies to the OS clipboard. Press <code>prefix m</code> again to return to native.
<code>WheelUpPane / WheelDownPane</code>	<code>copy-mode override</code>	While tmux mouse is ON, the wheel always scrolls tmux's own buffer — even inside Claude Code.
<code>allow-passthrough</code>	<code>on</code>	Apps can pass escape sequences through tmux.
<code>extended-keys</code>	<code>on</code>	Modified keys (e.g. Shift+Enter) reach the app.

Default: native terminal scrolling (mouse off)

The shipped config ships with **mouse off** as the default. With mouse off tmux emits **zero** mouse-tracking enables (no CSI `?1000h` / `?1002h` / `?1006h`), so the emulator's own mouse is unobstructed: scroll with the **wheel / trackpad / scrollbar exactly as**

in any terminal, and select text with a native click-drag → Cmd-C / right-click → Copy. This is the path that works everywhere — Linux and macOS, every emulator, inside or outside a TUI — proven at the wire level by test 59.

Scrolling tmux's own scrollbar inside a TUI (prefix m)

A full-screen TUI such as the Claude Code / HelixCode TUI runs on the terminal's **alternate screen**, which has no scrollbar of its own. With the default mouse off, your wheel scrolls the *terminal's* buffer, not tmux's 50 000-line history.

To page back through tmux's scrollbar inside such a TUI, press **prefix m** to toggle tmux mouse **ON**. The shipped config overrides WheelUpPane / WheelDownPane so that, while mouse is on, the wheel (and a touch-scroll, which terminals deliver as wheel events) **always** drives tmux copy-mode scrollbar regardless of what the running application does with the mouse. Press **prefix m** again to return to native terminal scrolling. The status line confirms the new state each time.

How to scroll

Default (mouse off) — native terminal scroll

- **Scroll up / down** — wheel, trackpad, or the terminal's scrollbar, exactly as in any other terminal window. tmux is not involved.

Inside a TUI, after prefix m (mouse on) — tmux scrollbar

- **Scroll up** — wheel up. tmux enters copy-mode and scrolls back.
- **Scroll down** — wheel down. When you reach the bottom, copy-mode exits automatically.
- Press **prefix m** again to return to native terminal scrolling.

Without a mouse — keyboard (and phones)

- Ctrl-b then [— enter copy-mode.
- Ctrl-b then PageUp — enter copy-mode **and** page up in one motion.
- In copy-mode: k/j line up/down, Ctrl-u/Ctrl-d half-page, PageUp/PageDown, g/G top/bottom, / search forward, ? search back, q to exit.

On a phone (Termux / Android)

- **Touch-scroll** the terminal — Termux delivers the swipe as wheel events, so it enters copy-mode and scrolls, exactly like a mouse.
- The Ctrl-b [route also works using Termux's extra-keys row (enable it with a long-press on the keyboard area if hidden).

Copying text

Default (mouse off): just select with the **native terminal** — click-drag (including across multiple lines), then `Cmd-C` / right-click → Copy. This works in every emulator on Linux and macOS, and it is the recommended copy path. To paste the OS clipboard back into a pane: native paste (`Cmd-V` / right-click → Paste), or prefix `P` to paste via a keyboard binding.

On demand (after prefix `m`, mouse on): a tmux mouse drag selects and copies on release; or in copy-mode press `v` to start a selection, move the cursor, then `y` or Enter to copy. The tmux selection is routed to your system clipboard automatically — the config detects `pbcopy` (macOS), `wl-copy` (Wayland), `xclip` (X11), or `termux-clipboard-set` (Termux) at copy time, and also emits OSC-52 so copying works over SSH and on terminals that support it.

Why the default flipped to `mouse off`: with `mouse on`, tmux emitted mouse-tracking enables that **suppressed** the emulator's native selection and right-click → Copy — the root cause of the long-standing "can't select / copy" reports. With `mouse off` the native mouse is unobstructed (proven by test 59), and prefix `m` gives you the tmux mouse on demand.

Verifying it works

The behaviour is covered by `scripts/tests/17_scrollback_copy_mode.sh`, which runs as part of the verification gate. It spawns a real session via `tmx new`, generates 3000 lines of output, proves the first line scrolled off the visible screen, then proves copy-mode can scroll back to it and copy it — all with captured runtime evidence per the anti-bluff covenant.

```
bash scripts/tests/17_scrollback_copy_mode.sh
# → PASS=13 FAIL=0 SKIP=0
```

Re-applying after a config change

`scripts/tmux.conf.template` is the single source. After editing it (or pulling an update), re-run `bash scripts/setup.sh` — or, in a running session, `tmx` picks up the template on the next `tmx new`. To reload into an already-running session: `Ctrl-b` then `:source-file ~/.tmux.conf`.

Sources verified 2026-06-13

- **tmux upstream man page** — <https://man.openbsd.org/tmux.1> (re-verified 2026-06-13 via WebFetch; OpenBSD ships the canonical upstream `tmux.1`). Confirms `set-clipboard` external semantics ("tmux will attempt to set the terminal clipboard but ignore attempts by applications to set tmux buffers" — the OSC-52 copy-out path) and that a flag/choice option set with its value omitted toggles its value (the prefix `m` / `set -g mouse` toggle). The `mouse off` ⇒ no mouse-tracking DECSET enables behaviour is proven at the wire

level in this repo by test 59 (`scripts/tests/59_*.sh`), which is the load-bearing authority for the default-architecture claims above.